

PROJECT

Design and Implementation of a VLAN-Based Network Using Static Routing, DHCP, and DNS in Cisco Packet Tracer

Objective

- To design a network topology consisting of two routers and multiple LAN segments using Cisco Packet Tracer.
- To implement VLAN segmentation on switches to logically separate network traffic.
- To configure inter-VLAN communication using the router-on-a-stick technique.
- To establish connectivity between routers using static routing.
- To configure DHCP services for automatic IP address allocation across all networks.
- To deploy a DNS server for domain name resolution within the network.
- To verify network connectivity using diagnostic commands such as `ping`, `tracert`, and `ipconfig`.
- To troubleshoot unreachable network issues by analyzing routing tables and host configurations.

Theory

1. Network Topology Design

A network topology defines how devices such as routers, switches, servers, and hosts are interconnected. In this project, a hierarchical topology was implemented where each router connects to a local switch and LAN, while the routers are interconnected through a point-to-point network. This design simulates communication between two separate network sites.

2. Virtual Local Area Network (VLAN)

A Virtual Local Area Network (VLAN) is a logical segmentation of a physical network that allows devices to be grouped into separate broadcast domains regardless of their physical location. VLANs improve network security, reduce broadcast traffic, and enhance network management.

In this project, VLAN 10 and VLAN 20 were configured on both switches to separate hosts into different logical networks.

3. Inter-VLAN Routing (Router-on-a-Stick)

Since devices in different VLANs cannot communicate directly, inter-VLAN routing is required. Router-on-a-Stick is a technique where a single physical router interface is divided into multiple subinterfaces, each configured with a VLAN ID and IP address. This allows the router to route traffic between VLANs.

In this project, subinterfaces were created on both routers to enable communication between VLAN 10 and VLAN 20.

4. Static Routing

Static routing is a routing method where routes are manually configured by a network administrator. It is simple, secure, and efficient for small networks. Static routes define the path that packets must take to reach remote networks.

In this project, static routes were configured on both routers to allow communication between networks located on different routers.

5. Default Routing

A default route is used when a router does not have a specific route for a destination network. It directs packets to a predefined next hop. Default routing simplifies routing tables and is commonly used in networks with a single exit point.

Although specific static routes were used in this project, the concept of default routing applies when forwarding unknown traffic to another router.

6. Dynamic Host Configuration Protocol (DHCP)

DHCP is a network protocol used to automatically assign IP addresses, subnet masks, default gateways, and DNS information to hosts. It reduces manual configuration errors and simplifies network administration.

In this project, DHCP pools were configured on routers to dynamically allocate IP addresses to all hosts in each VLAN.

7. Domain Name System (DNS)

DNS is a hierarchical naming system that translates domain names into IP addresses. It allows users to access network resources using easy-to-remember names instead of numerical IP addresses.

A DNS server was configured in this project to resolve domain names such as *www.test.com* into the corresponding IP address.

8. Network Verification and Troubleshooting

Network verification ensures that devices can communicate correctly and routing is functioning as expected. Common diagnostic tools include:

- **ping** — tests connectivity between devices
- **tracert** — identifies the path packets take
- **ipconfig** — displays host IP configuration
- **show ip route** — verifies routing tables

Troubleshooting involves identifying and resolving connectivity issues by analyzing device configurations and routing paths.

NETWORK CONFIGURATION

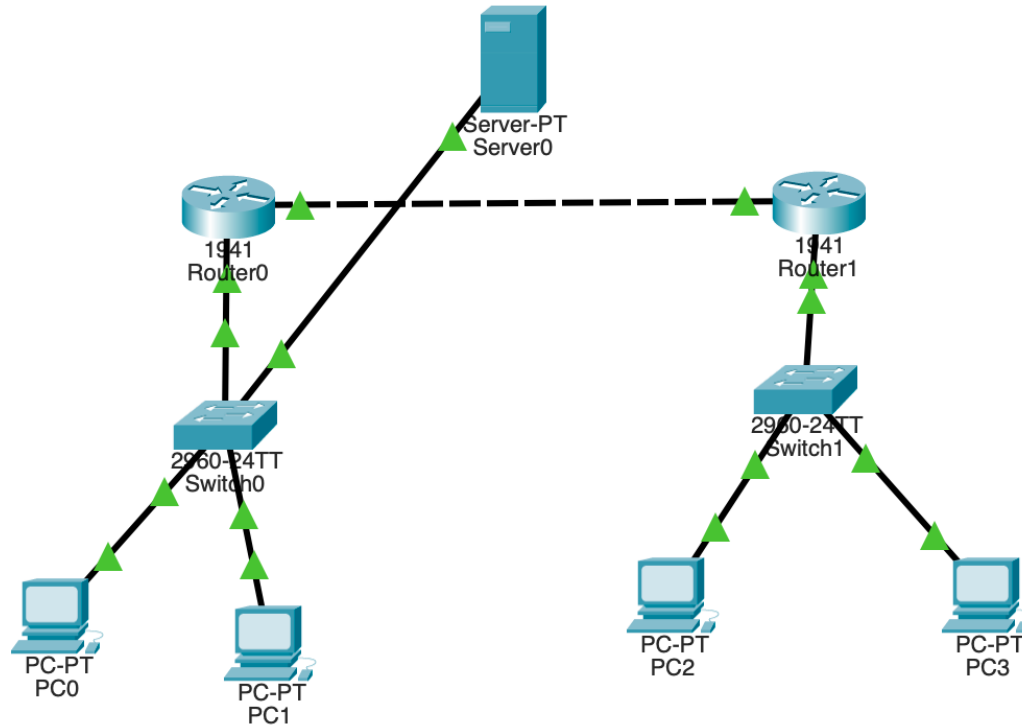


Fig: Network Configuration

Table 1. VLAN and Subnet Plan (Addressing Scheme)

Site	VLAN	Purpose	Network/Mask	Default Gateway (Router Subinterface)
Left (Switch0/Router0)	10	User LAN (PC0 + DNS Server)	192.168.10.0 /24	192.168.10.1 (R0 G0/0.10)
Left (Switch0/Router0)	20	User LAN (PC1)	192.168.20.0 /24	192.168.20.1 (R0 G0/0.20)
Right (Switch1/Router1)	10	User LAN (PC2)	192.168.110.0 /24	192.168.110.1 (R1 G0/0.10)
Right (Switch1/Router1)	20	User LAN (PC3)	192.168.120.0 /24	192.168.120.1 (R1 G0/0.20)
WAN	—	Router-to-router link	10.0.0.0 /30	R0=10.0.0.1, R1=10.0.0.2

Table 2. End Devices (PCs + Server) IP Configuration

Device	VLAN	IP Assignment	Observed	Subnet Mask	Default Gateway	DNS Server
PC0 (Left)	10	DHCP	192.168.10.21	255.255.255.0	192.168.10.1	192.168.10.10
PC1 (Left)	20	DHCP	192.168.20.21	255.255.255.0	192.168.20.1	192.168.10.10
PC2 (Right)	10	DHCP	192.168.110.21	255.255.255.0	192.168.110.1	192.168.10.10
PC3 (Right)	20	DHCP	192.168.120.21	255.255.255.0	192.168.120.1	192.168.10.10
Server0 (DNS)	10	Static	192.168.10.10	255.255.255.0	192.168.10.1	192.168.10.10

Table 3. Switch0 (2960) VLAN/Port Assignment (Left Site)

Switch	Port	Connected Device	Port Mode	VLAN
Switch0	Fa0/1	PC0	Access	10
Switch0	Fa0/2	PC1	Access	20
Switch0	Fa0/3	Router0 (G0/0)	Trunk	Allowed VLANs: 10,20
Switch0	Fa0/4	Server0	Access	10

Table 4. Switch1 (2960) VLAN/Port Assignment (Right Site)

Switch	Port	Connected Device	Port Mode	VLAN
Switch1	Fa0/1	PC2	Access	10
Switch1	Fa0/2	PC3	Access	20
Switch1	Fa0/3	Router1 (G0/0)	Trunk	Allowed VLANs: 10,20

Table 5. Router0 Interface Configuration (Left Router)

Router	Interface	Type	VLAN Tag	IP Address/Mask	Purpose
Router0	G0/0	Physical	—	(no IP)	Trunk link to Switch0
Router0	G0/0.10	Subinterface	10	192.168.10.1 /24	Gateway for VLAN10 (Left)
Router0	G0/0.20	Subinterface	20	192.168.20.1 /24	Gateway for VLAN20 (Left)
Router0	G0/1	Physical	—	10.0.0.1 /30	WAN link to Router1

Table 6. Router1 Interface Configuration (Right Router)

Router	Interface	Type	VLAN Tag	IP Address/Mask	Purpose
Router1	G0/0	Physical	—	(no IP)	Trunk link to Switch1
Router1	G0/0.10	Subinterface	10	192.168.110.1 /24	Gateway for VLAN10 (Right)
Router1	G0/0.20	Subinterface	20	192.168.120.1 /24	Gateway for VLAN20 (Right)
Router1	G0/1	Physical	—	10.0.0.2 /30	WAN link to Router0

Table 7. DHCP Configuration Summary (All Networks)

Router	DHCP Pool Name	Network	Default Router	DNS Server	Excluded Range (Reserved)
Router0	VLAN10_LEFT	192.168.10.0/24	192.168.10.1	192.168.10.10	192.168.10.1–192.168.10.20
Router0	VLAN20_LEFT	192.168.20.0/24	192.168.20.1	192.168.10.10	192.168.20.1–192.168.20.20
Router1	VLAN10_RIGHT	192.168.110.0/24	192.168.110.1	192.168.10.10	192.168.110.1–192.168.110.20
Router1	VLAN20_RIGHT	192.168.120.0/24	192.168.120.1	192.168.10.10	192.168.120.1–192.168.120.20

Table 8. Routing Configuration (Static Route)

Router	Destination Network	Mask	Next Hop	Route Type
Router0	192.168.110.0	255.255.255.0	10.0.0.2	Static
Router0	192.168.120.0	255.255.255.0	10.0.0.2	Static
Router1	192.168.10.0	255.255.255.0	10.0.0.1	Static
Router1	192.168.20.0	255.255.255.0	10.0.0.1	Static

Table 9. Default Route

Router	Default Route Command	Meaning
Router1	ip route 0.0.0.0 0.0.0.0 10.0.0.1	Send unknown traffic to Router0

```

Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.1 to network 0.0.0.0

 10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
   C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
   L       10.0.0.2/32 is directly connected, GigabitEthernet0/1
   S       192.168.10.0/24 [1/0] via 10.0.0.1
   S       192.168.20.0/24 [1/0] via 10.0.0.1
 192.168.110.0/24 is variably subnetted, 2 subnets, 2 masks
   C       192.168.110.0/24 is directly connected, GigabitEthernet0/0.10
   L       192.168.110.1/32 is directly connected, GigabitEthernet0/0.10
 192.168.120.0/24 is variably subnetted, 2 subnets, 2 masks
   C       192.168.120.0/24 is directly connected, GigabitEthernet0/0.20
   L       192.168.120.1/32 is directly connected, GigabitEthernet0/0.20
S*  0.0.0.0/0 [1/0] via 10.0.0.1

```

At end S* shows that a default route has been configured

Table 10. DNS Server Configuration

Component	Setting
DNS Server Device	Server0
IP Address	192.168.10.10/24
Default Gateway	192.168.10.1
DNS Service	ON
DNS Record (A)	www.test.com → 192.168.10.10
Verification	nslookup www.test.com, ping www.test.com

RESULT (PROOF TO THE CONFIGURTION)

VLAN Configuration

On Switch0

```
Switch#show vlan brief
VLAN Name                Status    Ports
-----
1    default                active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
10   VLAN10                  active    Fa0/1, Fa0/4
20   VLAN20                  active    Fa0/2
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
```

On Switch1

```
Switch#show vlan brief
VLAN Name                Status    Ports
-----
1    default                active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                           Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                           Fa0/24, Gig0/1, Gig0/2
10   VLAN10                  active    Fa0/1
20   VLAN20                  active    Fa0/2
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default          active
Switch#
```

Trunking

On Switch0

```
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/3     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/3     10,20

Port      Vlans allowed and active in management domain
Fa0/3     10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/3     10,20

Switch#
```

On Switch1

```
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/3     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/3     10,20

Port      Vlans allowed and active in management domain
Fa0/3     10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/3     10,20

Switch#
```

Router Interface Configuration

On Router0

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    unassigned      YES unset   up          up
GigabitEthernet0/0.10 192.168.10.1    YES manual   up          up
GigabitEthernet0/0.20 192.168.20.1    YES manual   up          up
GigabitEthernet0/1    10.0.0.1        YES manual   up          up
Vlan1           unassigned      YES unset   administratively down down
```

On Router1

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    unassigned      YES unset   up          up
GigabitEthernet0/0.10 192.168.110.1    YES manual   up          up
GigabitEthernet0/0.20 192.168.120.1    YES manual   up          up
GigabitEthernet0/1    10.0.0.2        YES manual   up          up
Vlan1           unassigned      YES unset   administratively down down
```


DHCP Working

On Router0

```
Router#show ip dhcp binding
IP address      Client-ID/      Lease expiration  Type
                Hardware address
192.168.10.21    00E0.A302.6AC4   --                Automatic
192.168.20.21    0000.0CCD.8667   --                Automatic
Router#
```

On Router1

```
Router#show ip dhcp binding
IP address      Client-ID/      Lease expiration  Type
                Hardware address
192.168.110.21   0040.0B60.AC76   --                Automatic
192.168.120.21   0009.7C9A.6584   --                Automatic
Router#
```

End-to-End Connectivity

On PC0 (Testing for PC2)

```
C:\>ping 192.168.110.21

Pinging 192.168.110.21 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.110.21: bytes=32 time=1ms TTL=126
Reply from 192.168.110.21: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.110.21:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

On PC3 (Testing for PC0)

```
C:\>ping 192.168.10.21

Pinging 192.168.10.21 with 32 bytes of data:

Reply from 192.168.10.21: bytes=32 time=1ms TTL=126
Reply from 192.168.10.21: bytes=32 time<1ms TTL=126
Reply from 192.168.10.21: bytes=32 time<1ms TTL=126
Reply from 192.168.10.21: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.10.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

On PC1(VLAN routing works)

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time=13ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms
```

DNS Resolution

On PC0

```
C:\>nslookup www.test.com

Server: [192.168.10.10]
Address: 192.168.10.10

Non-authoritative answer:
Name: www.test.com
Address: 192.168.10.10
```

```
C:\>ping www.test.com

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Troubleshooting

```
C:\>tracert 192.168.10.21

Tracing route to 192.168.10.21 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    192.168.20.1
  1  0 ms    0 ms    0 ms    192.168.10.21

Trace complete.
```

It shows packet path

Discussion

The project successfully demonstrated the design and implementation of a segmented network using VLANs, static routing, DHCP, and DNS services within a simulated Cisco Packet Tracer environment. VLAN configuration allowed logical separation of hosts into distinct broadcast domains, improving traffic management and network organization. Inter-VLAN communication was enabled using the router-on-a-stick approach, which effectively allowed devices in different VLANs to communicate through subinterfaces on the router. Static routing was used to establish communication between networks connected to different routers, illustrating how manually configured routes can control packet forwarding paths in small to medium-sized networks.

The implementation of DHCP simplified host configuration by dynamically assigning IP addresses, default gateways, and DNS information, reducing the possibility of manual configuration errors. DNS services enabled name resolution, allowing hosts to communicate using domain names rather than numerical IP addresses. During the verification phase, network diagnostic tools such as ping, traceroute, and routing table inspection were used to confirm connectivity and validate the correctness of configurations. Additionally, troubleshooting steps were performed to resolve connectivity issues caused by DHCP address changes, demonstrating the importance of verifying host configurations and routing paths when diagnosing network problems. Overall, the project provided practical insight into fundamental networking concepts and their real-world applications.

Conclusion

This project achieved its objective of designing and implementing a functional multi-network topology using Cisco Packet Tracer. VLAN segmentation successfully separated network traffic, while inter-VLAN routing enabled communication between logically distinct networks. Static routing provided connectivity between routers, ensuring seamless communication across different sites. The deployment of DHCP automated IP address allocation, and the DNS server facilitated efficient domain name resolution within the network.

Through systematic verification and troubleshooting, the network was confirmed to operate as intended, with successful end-to-end connectivity across all devices. The project highlighted the importance of proper network design, addressing schemes, and routing configuration in ensuring reliable communication. It also reinforced practical skills in diagnosing and resolving network issues using standard diagnostic tools. Overall, the implementation demonstrates how core networking technologies can be integrated to create a scalable, efficient, and manageable network infrastructure.

